

Exam Date & Time: 11-Jan-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - JANUARY-2024

Fundamentals of Machine Learning [CSE-3173]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain the difference between correlation and regression using example. (2)
- A)
- B) Design a machine learning model for stock market prediction using multiple linear regression by mentioning feature variables, target variable, model setup, error function and model training and evaluation. (5)
- C) Discuss different error functions in linear regression technique. (3)
- 2) Explain how imbalanced data set are handled using K-Nearest Neighbour. [2 Marks].

- Given a dataset of fruits with two features: weight (in grams) and sweetness score (ranging from 1 to 10), and the class labels are "Apple" and "Orange." Classify a new fruit with a weight of 120 grams and a sweetness score of 7 using a KNN classifier with $k=3$. [3 Marks].

Weight (in grams)	Sweetness score	Class
110	8	Apple
120	6	Apple
130	7	Orange
150	3	Orange
110	5	Apple

(5)

- B) Consider below dataset of student's performance in a university. Create new decision tree with the help of information gain for each candidate attribute at each step in growing the tree for a student performance dataset: (5)

Hours	Previous Scores	Attendance	Performance
5	85	90	Pass
4	78	95	Pass
6	92	80	Pass
3	72	75	Fail
5	88	88	Pass

- 3) With the help of code snippet illustrate classification techniques using Support Vector Machines algorithm for non-linear data set using suitable example. (5)

A)

- B) Consider a dataset of text messages and their corresponding labels Spam or Not Spam. classify the new text and predict the class using Naive Bayes:

Text	Class
Buy cheap luxury watches now!	Spam
Meet me at the park later.	Not Spam
Win a free vacation today!	Spam
Can we reschedule our meeting?	Not Spam
Congrats, you've won a prize!	Spam

(5)

Given a new message (e.g., " Meeting tomorrow "). Classify the new message and predict the class using Naive Bayes.

- 4) Explain Lasso regression technique suitable dataset and discuss cost function and evaluation methods. (5)

A)

- B) Explain Extreme gradient classifier using suitable dataset and discuss algorithms with block diagram. (5)

- 5) Explain generalization and various related issues in machine learning model. (3)

A)

- B) Explain the collaborative filter based recommender systems with use case and advantages. (3)

- C) Explain the collaborative filter using matrix factorization technique (4)

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